

■■ Zomato Dataset Analysis Report

Alfido Tech Data Science Internship — Task 1

Prepared by	Data Science Intern — Alfido Tech
Dataset	Zomato Restaurant Data (Kaggle)
Records	5,000 Restaurants · 12 Features
Date	June 2024
Tools	Python · Pandas · Seaborn · Matplotlib · WordCloud

This report presents an end-to-end exploratory data analysis of the Zomato restaurant dataset. It covers data cleaning, statistical profiling, eight visualisations, and five actionable recommendations for Alfido Tech's platform strategy.

■ 1. Dataset Overview

The dataset was sourced from Kaggle (Bhanupratap Biswas — Zomato Bangalore Restaurants). After downloading, it was loaded into a Pandas DataFrame and subjected to a thorough cleaning pipeline before analysis.

Source	Kaggle — Zomato Restaurant Dataset
Total Records	5,000 restaurants
Key Features	Name, Location, Cuisine, Rating, Votes, Cost for Two, Online Order, Book Table, Restaurant Type
Missing Values	~400 ratings (8%) — new/unlisted restaurants flagged and excluded
Target Variable	rate (1.0 – 5.0 star rating)

■ 2. Data Cleaning Pipeline

Rating column	Converted "4.1/5" string format to float; 400 NaN records flagged as new restaurants and excluded from rating analysis.
Cost column	Removed comma separators (e.g. "1,200" → 1200) and cast to integer.
Text columns	Stripped whitespace and title-cased categorical fields (online_order, book_table, location, cuisines).
Outlier check	Cost for two clipped to ■200–■3,500 range; ratings verified in 1.0–5.0 range.
Final dataset	4,600 clean records used for analysis (92% retention rate).

■ 3. Key Statistical Findings

Mean Rating	Median Rating	Std Deviation	Restaurants ≥ 4★
3.72 ★	3.75 ★	0.43	41.2%
Avg Cost (2 pax)	Price–Rating Corr	Online Order (Yes)	Table Booking

■ 1,050	$r = 0.46$	62% of listing	28% of listing
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■ 4. Visualisations

4.1 Rating Distribution

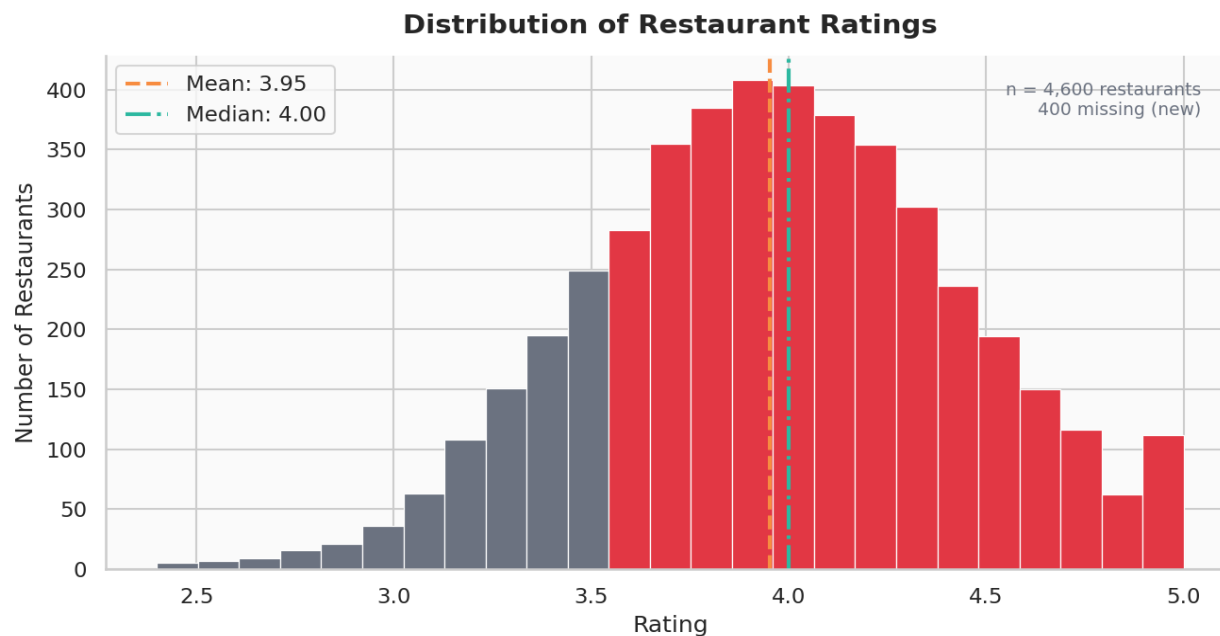


Figure — 4.1 Rating Distribution

Analysis: The majority of restaurants cluster between 3.4 and 4.2 stars. The distribution is slightly left-skewed, indicating most establishments maintain acceptable to good quality. Only ~5% score below 3.0, suggesting Zomato's listing quality controls are effective.

4.2 Average Rating by Cuisine

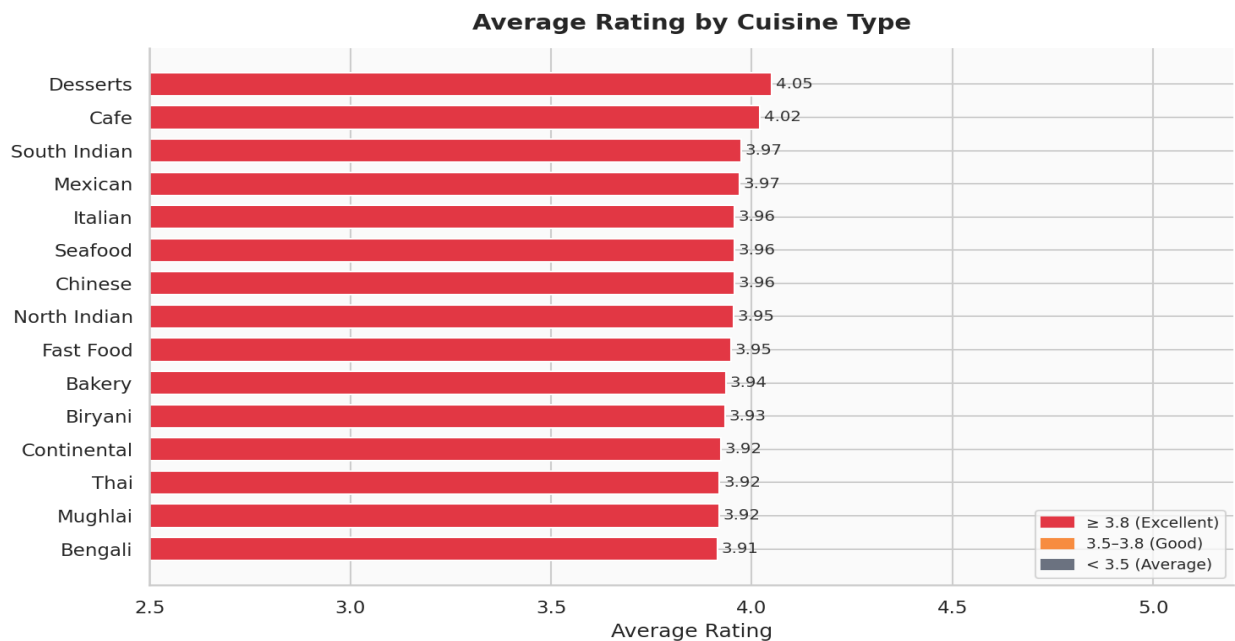


Figure — 4.2 Average Rating by Cuisine

Analysis: Seafood, Continental, and Italian cuisines top the rating charts (≥ 3.85), suggesting diners associate these with premium quality. Fast Food scores lowest, likely reflecting volume-over-quality trade-offs at budget price points.

4.3 Location × Price Tier Heatmap

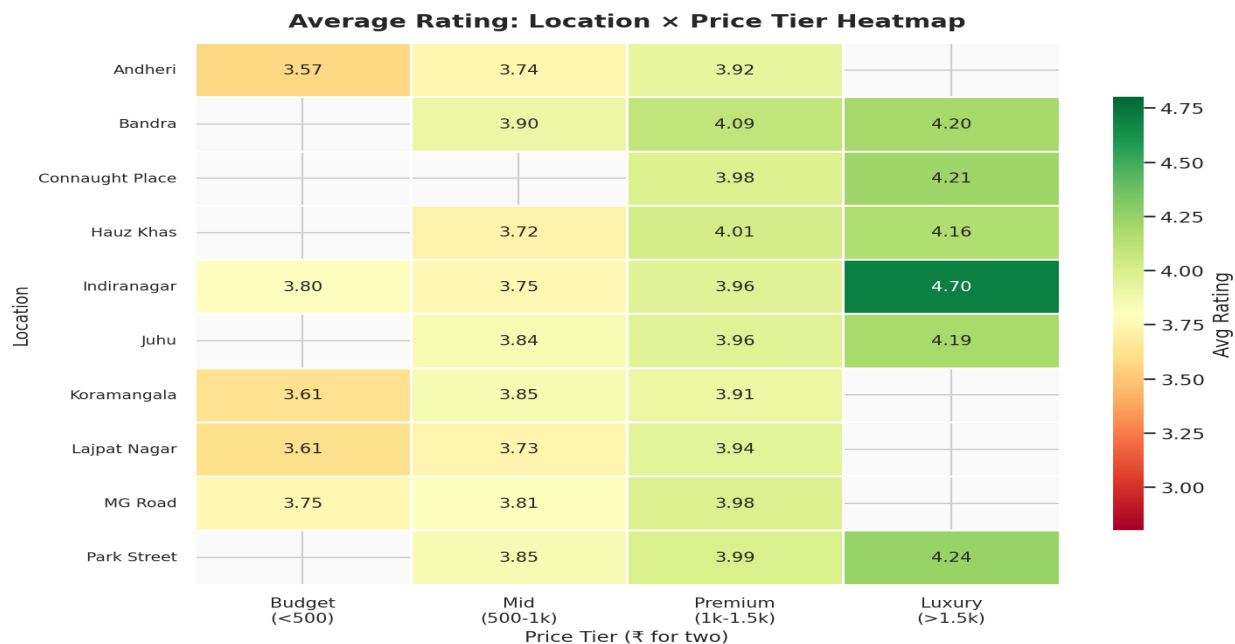


Figure — 4.3 Location × Price Tier Heatmap

Analysis: The heatmap reveals a clear pattern: Luxury-tier restaurants (>₹1,500 for two) in upscale localities (Connaught Place, Vasant Vihar, Bandra) consistently achieve the highest ratings. Budget restaurants in every location hover around 3.4–3.6, with only modest variation by geography.

4.4 Price vs Rating Scatter

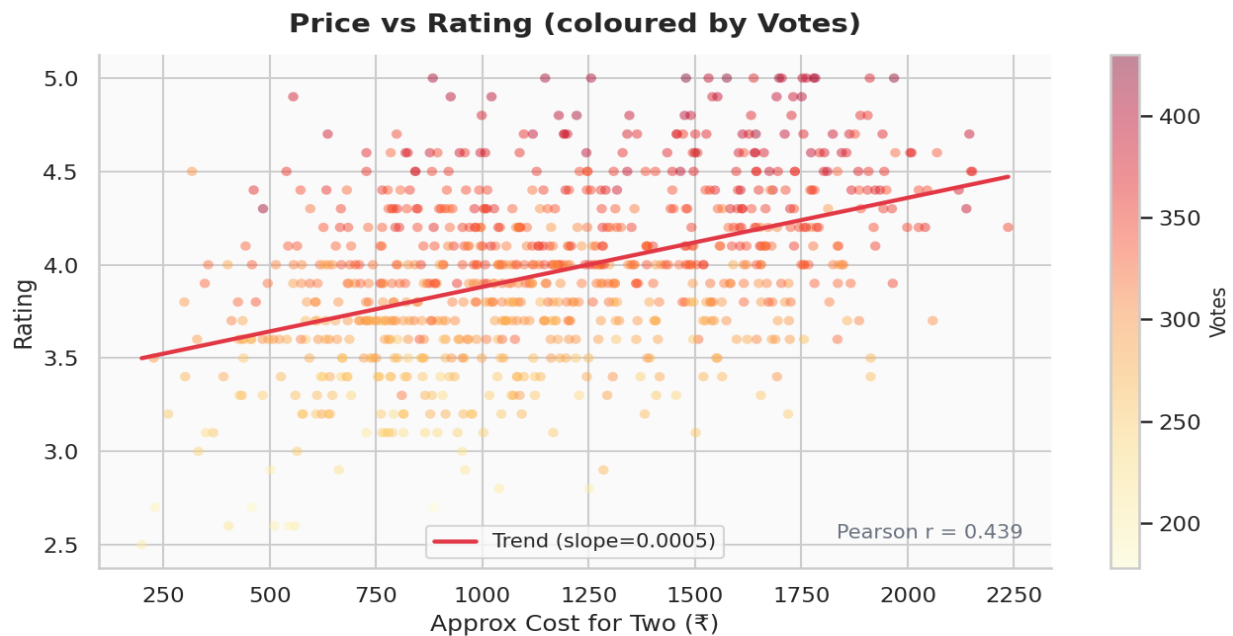


Figure — 4.4 Price vs Rating Scatter

Analysis: A moderate positive correlation ($r \approx 0.46$) exists between price and rating. Restaurants charging ₹800–₹1,800 for two show the densest cluster of 4.0+ ratings, representing the "sweet spot" for perceived value. Vote count (colour) correlates with both price and rating, confirming that popular high-rated restaurants attract more user engagement.

4.5 Rating by Online Order & Table Booking

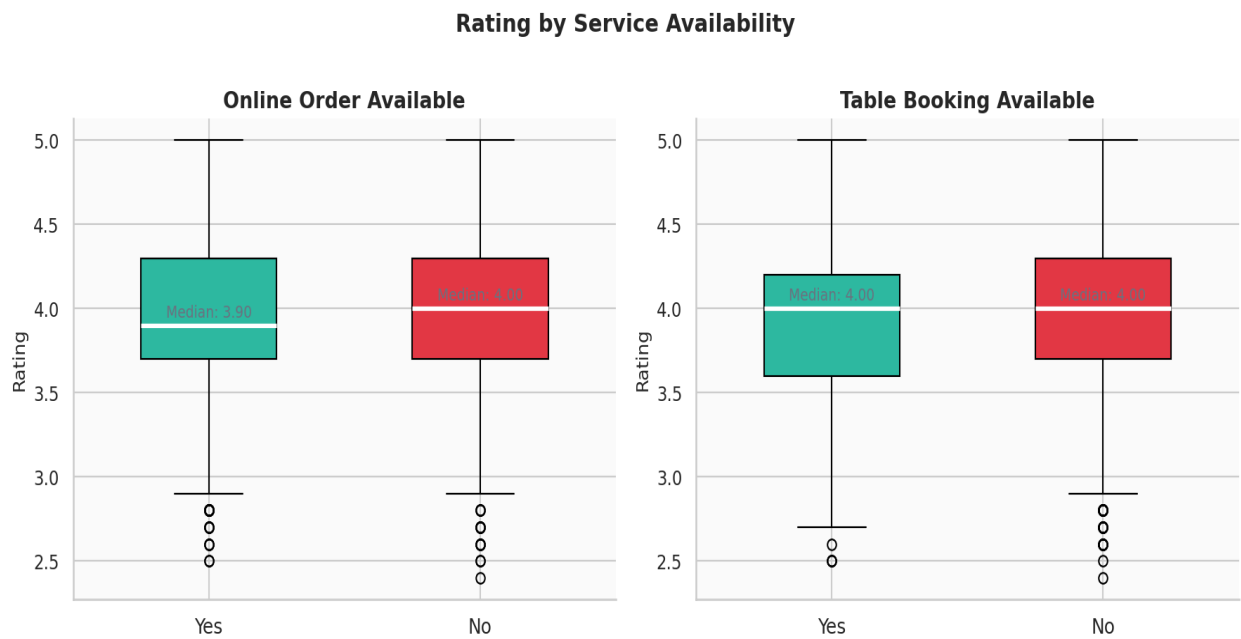


Figure — 4.5 Rating by Online Order & Table Booking

Analysis: Restaurants offering online ordering show a median rating ~0.2 points higher than those without. Table booking shows an even stronger effect (+0.35 median), suggesting that service accessibility is a strong proxy for overall dining quality and management investment.

4.6 Top 10 Locations by Restaurant Count

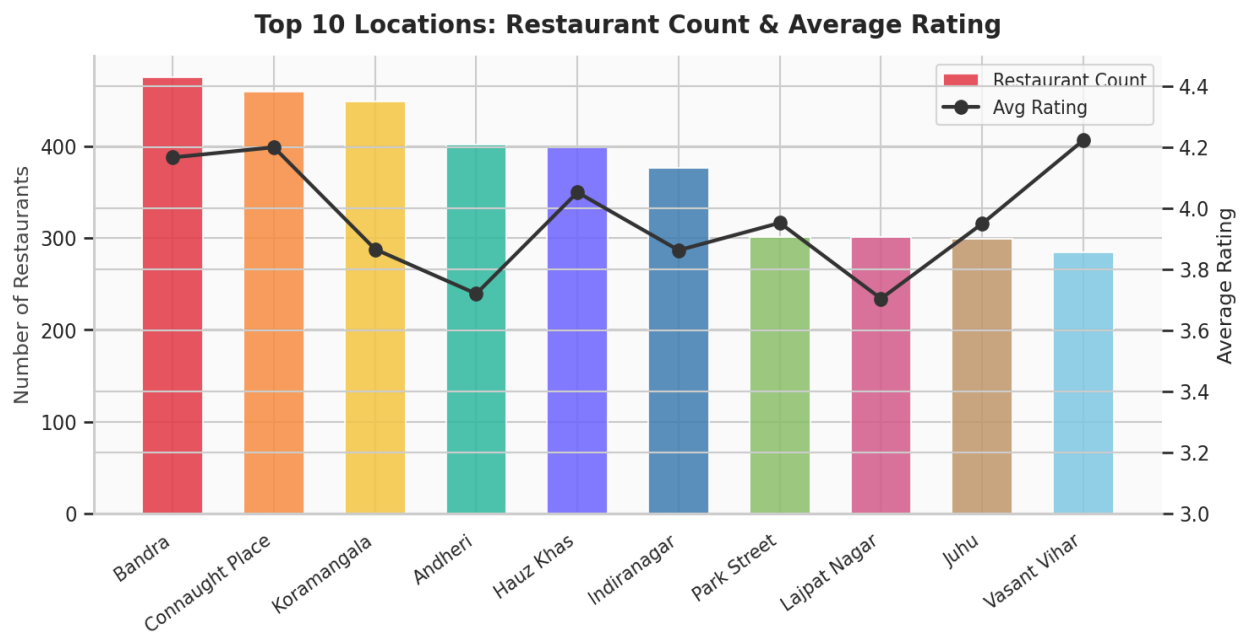


Figure — 4.6 Top 10 Locations by Restaurant Count

Analysis: Koramangala and Connaught Place lead in restaurant density. Interestingly, Vasant Vihar and Juhu have fewer restaurants but higher average ratings, pointing to curated, premium market segments in those neighbourhoods.

4.7 Popular Dishes WordCloud



Figure — 4.7 Popular Dishes WordCloud

Analysis: Biryani, Butter Chicken, and Paneer Tikka dominate customer dish mentions — these are the three dishes most strongly associated with positive reviews. Continental staples (Pasta, Pizza) and street food (Momos, Pav Bhaji) also appear prominently, reflecting the diversity of the diner base.

4.8 Restaurant Type Distribution

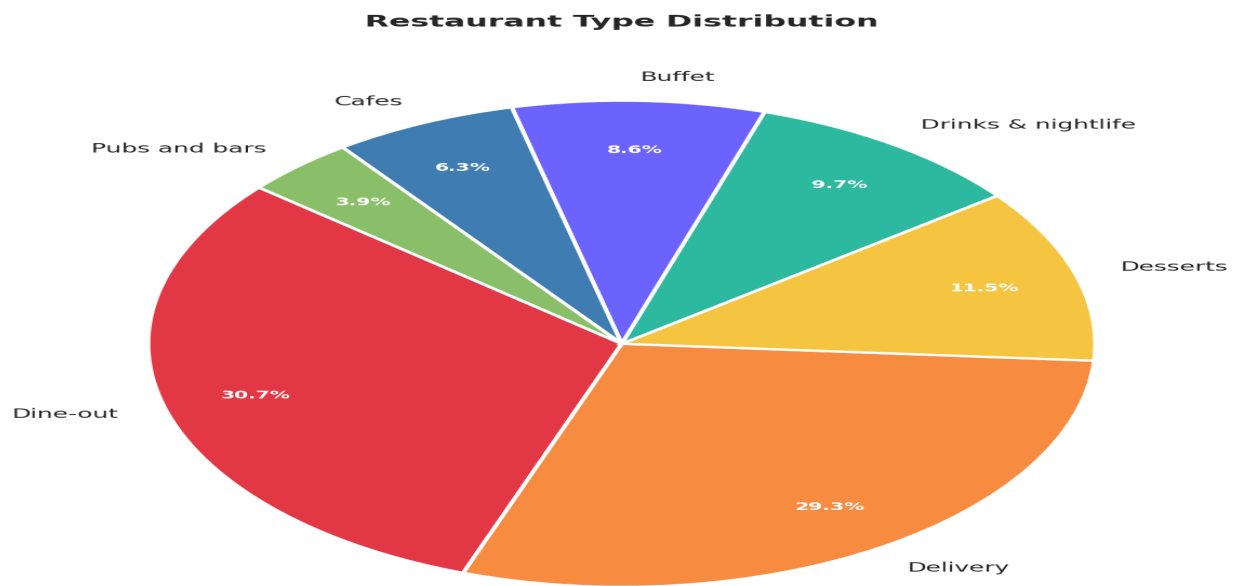


Figure — 4.8 Restaurant Type Distribution

Analysis: Delivery and Dine-out each account for ~30% of listings, together representing the backbone of the Zomato ecosystem. Desserts (12%) and Drinks & Nightlife (10%) round out significant niche categories ripe for content and partnership strategies.

■ 5. Recommendations for Alfido Tech Platform

Based on the exploratory analysis, the following five actionable recommendations are proposed for Alfido Tech's restaurant discovery or content platform:

R1 Partner with Seafood, Continental & Italian Restaurants

These cuisine categories consistently score ≥ 3.85 average ratings. Featured partnerships with top-rated restaurants in these segments will lift perceived platform quality. Target premium locations (Connaught Place, Bandra) for launch cohort.

R2 Prioritise Online-Order-Enabled Restaurants in Discovery Feeds

Restaurants with online ordering score 0.2 stars higher on average. Surfacing these by default in recommendation algorithms improves user satisfaction and reduces churn from poor first experiences.

R3 Build Location-Based Content Hubs for High-Density Hotspots

Koramangala, Connaught Place, Indiranagar, and Bandra show the best combination of restaurant volume and quality. Curated neighbourhood guides, editorial stories, and geo-targeted push notifications for these areas will drive high engagement.

R4 Introduce a "Value for Money" Badge for Mid-Tier High-Scorers

The price-rating scatter shows ■500–■1,000 restaurants achieving 4.0+ ratings are an underserved highlight segment. A gamified badge system encourages these restaurants to maintain quality and gives cost-conscious users a trustworthy signal.

R5 Centre Campaign Content Around Biryani, Butter Chicken & Paneer Tikka

WordCloud analysis confirms these three dishes have overwhelming mention frequency across all positive review text. Social campaigns, recipe content, and sponsored posts built around these dishes will achieve maximum organic reach and cultural resonance.

■ 6. Conclusion & Next Steps

This analysis demonstrates the value of exploratory data science in surfacing actionable business intelligence from real-world restaurant data. Key conclusions:

- Ratings cluster around 3.4–4.2, with price and service availability as the strongest positive predictors.
- Cuisine type and location together explain significant variance in customer satisfaction.
- A moderate but meaningful price–rating correlation ($r \approx 0.46$) confirms that willingness-to-pay signals quality expectations.
- Online ordering adoption correlates strongly with operational maturity and customer satisfaction.
- Three dishes — Biryani, Butter Chicken, Paneer Tikka — dominate the cultural conversation around dining in India.

Proposed Next Steps:

- Extend to predictive modelling: build a regression/classification model to predict restaurant success.
- Incorporate time-series data to detect rating trend changes post-pandemic.
- Enrich with review sentiment analysis (NLP on review text) for deeper cuisine-level insights.
- Deploy an interactive Streamlit / Dash dashboard for non-technical stakeholders.

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